

Issues for temporal counterpart theory

Handout for *Time and Modality* seminar

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1 Time travel

“Portal travel” in Neo-Newtonian spacetime.

Standard worm-theoretic account.

Challenges for the B-theorist:

- In cases of self-meeting, what is truth value of (e.g.) ‘Bill is sitting at t' when ‘ t' ’ is assigned to a slice that contains Bill twice over?
- What is going on with present-tense ‘This person is/is not sitting right now’ as spoken by someone who doesn’t know there is time travel going on? What about when spoken by someone who does know? Does it change if the speaker is the time traveller, using ‘I’ rather than ‘this person’?
- What about ‘In one hour, Jim will be walking on an oolitic reef?’ What about ‘One hour ago, Maria was freezing, and one hour ago, Zack was sweltering, so one hour ago, Maria was colder than Zack’?

The obvious temporal counterpart theory account: trace temporal counterpart relation via portals.

Surprises and challenges for this view:

- You can’t visit your great-grandchildren, Socrates, ancient Greece, etc.—only their temporal counterparts.
- Unless you find portals with *exactly* the right spacing, you’ll never see home and friends again.
- There really are two people involved in putative cases of “meeting your past self”, just as onlookers think.
- It’s not really *time* travel - just spacetime travel.

Alternative, contextualist temporal counterpart theory account: trace temporal counterpart relation for one salient object via portals; key everything else to that.

2 Special relativity

View one: Counterparts go by proper time along each worm.

- Problem of never meeting the same people again.

View two: Overpopulation: a distinct ordinary object for every flat worm-slice. Counterparts are parallel. Ordinary relations like *friendship* hold between slice-mates.

- Is this kind of population explosion really credible?
- Is it really in the spirit of relativity to say that I have an epistemically inaccessible

'true velocity' (i.e my velocity relative to the frame in which *I* am perpendicular to the time axis)?

- How should I form credences about my true velocity?
- Do people with a high true velocity make rampant errors about shape, duration, simultaneity, temporal order. . . ?
- How to extend to general relativity?

View three: Ultra-vagueness: 'There is one slicing such that all people and other ordinary objects occupy leaves in that slicing' is definitely true, but '*x* is a leaf of the slicing that contains ordinary objects' is never definitely true or definitely false.

- Background: classical-theorem-preserving theories of vagueness.
- What are the vague words? They include nouns like 'person', and pronouns like 'I'. Why some philosophers don't like the idea that 'I' is vague in this way.
- View three fits neatly with a points-only ontology, unlike views one and two.
- Will the vagueness in esoteric questions like 'What is our true velocity?' infect everyday questions about shape, duration, simultaneity, and temporal order with an excessive amount of vagueness?

View four: Crazy context-sensitivity. Everyone speaks the determinate truth when they say 'My current true velocity is zero, and other people's true velocity is their velocity relative to me'.

- When my son says 'My father is a philosopher', he doesn't refer to me. . . .

View four: Vagueness with contextual narrowing down.

- Background: the association of vagueness with context-sensitivity.
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